Project Overview

This project is a mobile-first website designed to present individual cross country racers' information in a format optimized for mobile screens. The goal was to create a responsive, accessible interface that allows users to easily explore athlete information, browse race history, and understand performance trends through visualizations. The site prioritizes usability on mobile devices by using a single-column layout, touch-friendly controls, and clearly organized information sections. Key athlete information remains visible through a sticky header, while race results are presented as expandable cards that allow users to access additional details when tapped. Accessibility and user preferences were an important part of the design. The site supports dark mode, high contrast mode, reduced motion preferences, and semantic landmark structure to ensure compatibility with assistive technologies.

Mobile-First Design

- The website was designed using a mobile-first approach. All base styles were written to support mobile devices first, using a single-column layout optimized for small screens.
- Content is vertically stacked so users can scroll naturally through athlete information, performance visualization, and race results without horizontal scrolling.
- Touch-friendly buttons and controls were used throughout the site to support mobile interaction. Interactive elements such as "More Info" buttons are large enough to be easily tapped.
- The athlete profile section uses a sticky header so the athlete's name and key information remain visible while the user scrolls through race data.

- Race results are displayed as expandable cards rather than a large table. This improves readability on mobile devices and reduces visual clutter while still allowing access to detailed information when needed.

CSS Quality

- CSS variables were used to manage colors and theme values. This improves maintainability and allows dark mode and high contrast mode to be implemented efficiently.
- Styling follows an 8-point spacing system to maintain consistent spacing across the layout. Margins, padding, and gaps are all based on multiples of 8px.
- The stylesheet is organized into clearly labeled sections including color modes, layout structure, header styling, card styling, and accessibility features.
- Repetition was minimized by using shared styles for cards, borders, and interactive elements.
- Modern layout techniques such as Flexbox and CSS Grid were used to create flexible and responsive layout structures.

Data Highlighting and Collapsible Sections

- Each race card contains a “More Info” button that expands to reveal detailed race data such as date, time, grade, and overall placement.
- This collapsible design keeps the mobile interface clean while still allowing users to access detailed data when desired.

- ARIA attributes such as aria-expanded and aria-controls were used to improve accessibility for these expandable sections.

Dynamic Visualizations

- A race performance visualization was created using Chart.js to display race times over time.
- This chart provides a clear visual representation of the athlete's performance progression across multiple seasons.
- The visualization dynamically reads race data and converts race times into a time-based line graph.

Sticky and Fixed Elements

- The athlete profile information is displayed inside a sticky header that remains visible while the user scrolls through race results.
- This ensures the athlete's identity and team information remain visible throughout the browsing experience.
- A floating toggle button was implemented in the lower corner of the screen to allow users to switch between light and dark modes.
- A separate floating button allows users to enable high contrast mode for improved accessibility.

Effects and Transitions

- Subtle interaction animations were added to improve user feedback and usability.

- Buttons include a small press animation that slightly scales the button when tapped, giving clear feedback that the interaction occurred.
- Race cards animate into view with a smooth fade and upward movement when they appear on the page.
- The dark mode toggle includes a sliding animation that moves between the sun and moon icons to indicate the current theme.

Custom Typography and Iconography

- The site uses simple, readable system fonts such as Arial and Helvetica to ensure compatibility across devices and maintain readability on smaller screens.
- Icons such as a sun and moon were used in the theme toggle to visually represent light and dark mode.
- Icons for cross country, school year, bio, and awards were used.
- Typography hierarchy was created using clear heading sizes to distinguish athlete information, section titles, and race data.

Dark Mode / High Contrast Mode

- Dark mode was implemented using CSS variables that change color values when dark mode is enabled.
- The site also detects system preferences using the prefers-color-scheme media query so that the interface can automatically match the user's device settings.
- A manual toggle allows users to switch between light and dark themes at any time.
- A high contrast mode was also implemented for users who require stronger visual clarity.

- High contrast mode increases border visibility, simplifies colors, and improves focus outlines for keyboard navigation.

Image Optimization

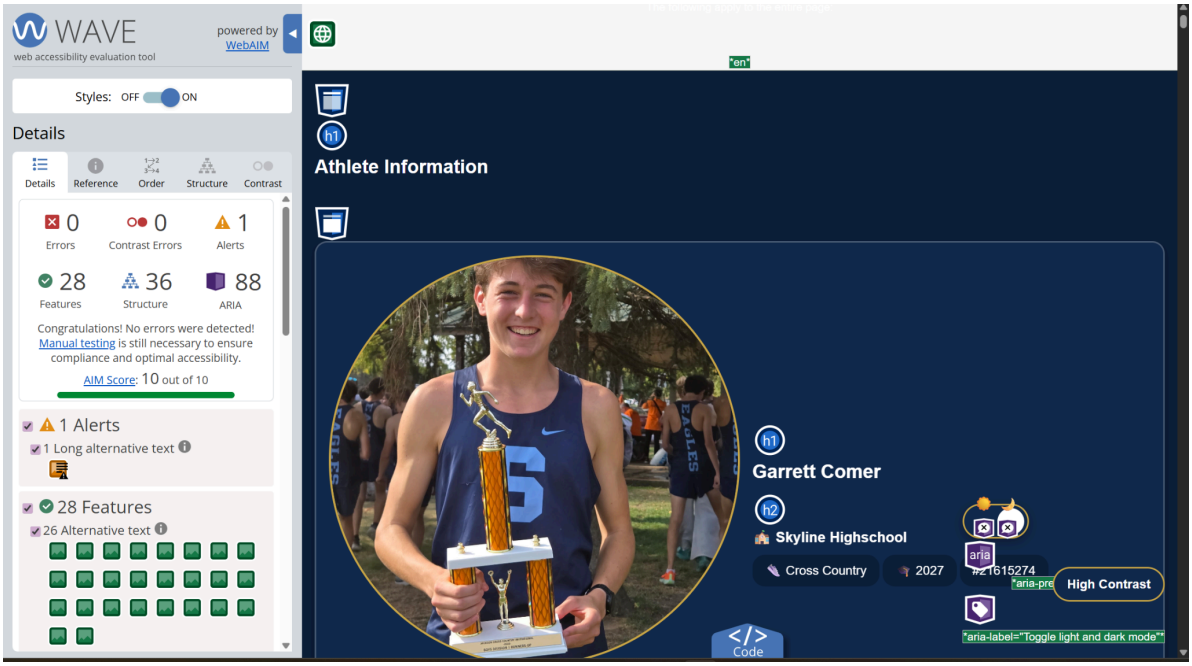
- The athlete profile image is displayed as circular images using border-radius and object-fit to ensure images remain properly cropped across devices.
- Race images are contained within responsive image containers that prevent distortion while maintaining the correct aspect ratio.
- Images scale proportionally to fit the layout without breaking the mobile design.

Animations

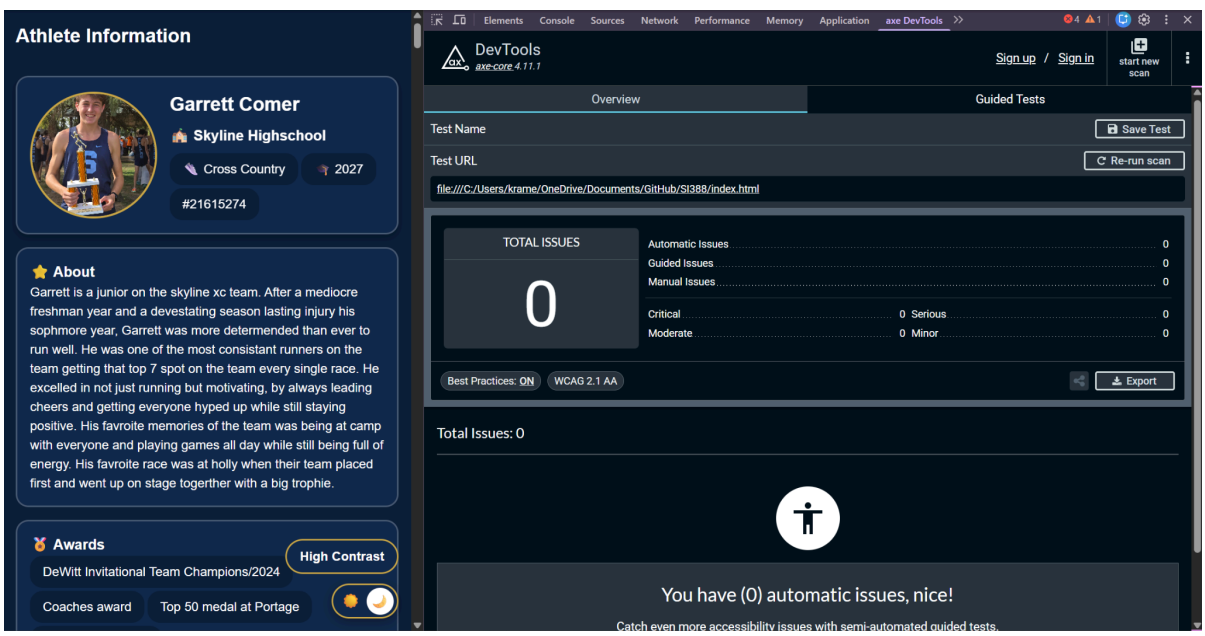
- Race cards animate into view using a fade-up animation when they load.
- Button press interactions provide visual feedback through a small scale animation.
- The dark mode toggle uses a sliding animation to clearly communicate theme changes.
- Animations were designed to be subtle and performance-friendly.

Accessibility Validation

All accessibility issues identified during testing were resolved.



Site passes WAVE Accessibility Test.



Site passes aXE DevTools Test.

Prefers Reduced Motion

- The prefers-reduced-motion media query was implemented to disable animations for users who prefer reduced motion.
- When this setting is enabled, animations and transitions are removed to prevent motion discomfort.

Reacts to User Color Schemes

- The site automatically adapts to the user's system color preference using the prefers-color-scheme media query.
- Users can also manually toggle light or dark mode using the floating theme toggle.

Overall Presentation and Clarity

- The website is organized into clear sections including athlete information, biography, performance visualization, and race results.
- Card-based layouts and consistent spacing make the interface easy to scan and navigate on mobile devices.
- The codebase is structured and commented to make the project easy to understand and maintain.
- Comments were made on the "style.css" sheet to indicate source code.